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1.5.3 Methods of Reproduction in Fungi

Asexual Reproduction

1. Zoospores: They are flagellate structures produced in zoosporangia (Example: Chytrids)

2. Conidia: The spores produced on condiophores (Example: *Aspergillus*)

3. Oidia/Thallospores/Arthrospores: The hypha divide and develop in to spores called oidia (Example: *Erysiphe*).

4. Fission: The vegetative cell divide into 2 daughter cells. (Example: *Schizosaccharomyces*-yeast).

5. Budding: A small outgrowth is developed on parent cell, which gets detached and become independent. (Example: *Saccharomyces*-yeast)

6. Chlamydospore: Thick walled resting spores are called chlamydospores (Example: *Fusarium*).

Sexual Reproduction

1. Planogametic copulation: Fusion of motile gamete is called planogametic copulation. a. Isogamy – Fusion of morphologically and physiologicall similar gametes. (Example: *Synchytrium)*. b. Anisogamy – Fusion of morphologically or physiologically dissimilar gametes (Example: *Allomyces*).

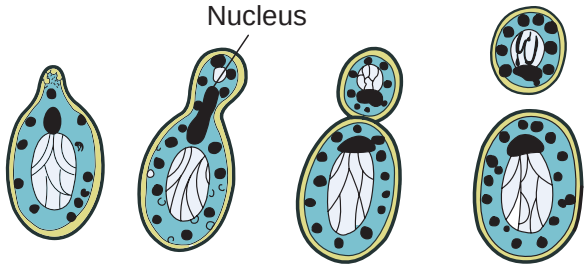
c. Oogamy – Fusion of both morphologically and physiologically dissimilar gametes. (Example: *Monoblepharis*)

2. Gametangial contact: During sexual reproduction a contact is established between antheridium and Oogonium (Example: *Albugo*)

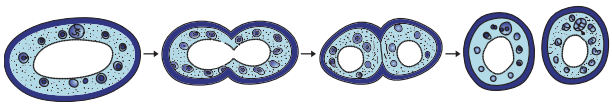
3. Gametangial copulation: Fusion of gametangia to form zygospore (Example: *Mucor, Rhizopus*).

4. Spermatization: In this method a uninucleate pycniospore/microconidium is transferred to receptive hyphal cell (Example: *Puccinia/Neurospora*)

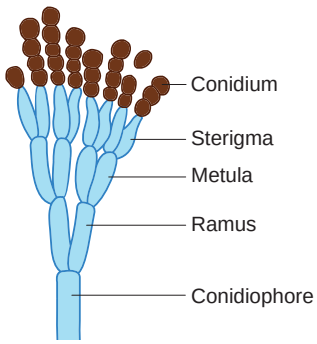
5. Somatogamy: Fusion of two somatic cells of the hyphae (Example: *Agaricus*)



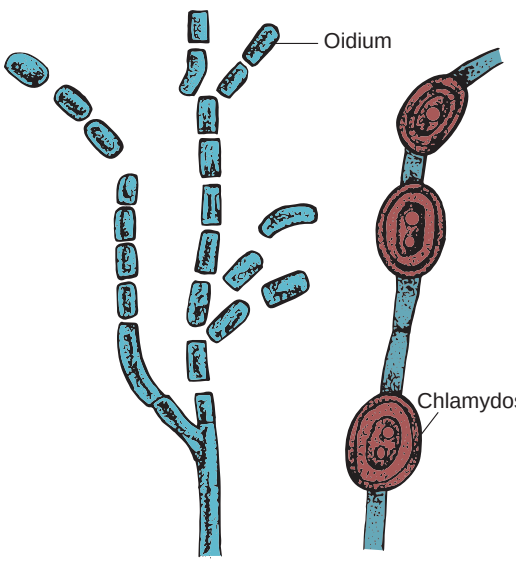
(a) Budding - Yeast



(b) Fission - Yeast



(c) Conidia formation - *Penicillium*



(d) Thallospore - *Erysiphe*

(e) Chlamydospore - *Fusarium*